RetinaCare Research Screening Report

RESEARCH PROTOTYPE - NOT A CLINICAL DOCUMENT. For investigational screening only; not a diagnosis. A licensed ophthalmologist must examine the patient.

1. Identification

Patient code: SYN-P4-001
Patient name: Synthetic Explain Test
Examination: EXAM-001
Eye: OD (right)
Examination date: 2026-10-28T15:51:00Z
Source image: OD-1.png (640 x 640 px)
Report generated: 2026-10-28 16:05:11Z

2. Status

COMPLETE - Stage I, Stage II (5 masks) and Stage III five-fold ensemble

3. Results

Outcome: Five-fold ensemble severity grade 3 (research only)
FIVE-FOLD ENSEMBLE (LG-DRG, official release v1.0) - RESEARCH ONLY. Not a clinical grade; licensed-image acceptance and clinical validation are pending.
Stage I screening gate: score 0.9983 (not a calibrated probability), threshold 0.2392, raw logit 6.386, route: continue cascade.
Stage II microaneurysms (MA): 58 foreground pixels of 640 x 640 (0.014% of the frame).
Stage II hemorrhages (HE): 1574 foreground pixels of 640 x 640 (0.384% of the frame).
Stage II exudates (EX): 8997 foreground pixels of 640 x 640 (2.197% of the frame).
Stage II cotton-wool spots (CWS): 400 foreground pixels of 640 x 640 (0.098% of the frame).
Stage II vessels (SwinHRUNet++): 47521 foreground pixels of 640 x 640 (11.602% of the frame).
Stage III LG-DRG (five-fold ensemble): technical severity grade 3; gradable probability 1.000.

4. Provenance

Stage I model: hierarchiretina-stage1-convnextv2l-768 sha256 9ad51e1f428dc59ab49f1f49e8f68a84cd23d8fd4c2c461b45b06bcc726880ac (CPU)
Stage II microaneurysms (MA) model sha256 ac874bee868bbd7109d1358deaa35ab59f38d7664a4ea7eec22e00c8be79aac5 (CPU)
Stage II hemorrhages (HE) model sha256 16f767a8deb3751962d00abfaa5aca27fb40b43c64a77108c78610111260b966 (CPU)
Stage II exudates (EX) model sha256 2aa92099f2333e3d1afe862a618a8436023bc03f14b5a8f62ea67debf2612207 (CPU)
Stage II cotton-wool spots (CWS) model sha256 6c928a75b28884ca19a58fa11daa6020868af57081b07b347961ce7b4c967882 (CPU)
Stage II vessels (SwinHRUNet++) model sha256 cc7ef0bb23732c7a1c89954244df9b6a5523f516c3818821811c667d2d5d76e0 (CPU)
Stage III model sha256 31f20d62f972a0a8f9119162c8a30b7e2a96509933f0a9bc9a3845b3a6129cb7, 3f0deb84cb04a6c301cf73efdd154fe18297252430e7c3afa36d2ab087092620, a55301c7e27251047e10211dfb10ca7f73674c826d89c4e7a772f3476df0a3ab, ee7c16614859ef3c8e8d1c8fbcc5a07388e3bd02622f74bdedcd8c4772b6fcfe, 659e65fb336342b2afc52385f4821a673e1b569b81454798820f4798fe876238 (CPU)
Source image sha256: 32d756fc923262daa5296c6852c1569e01394b37db57e72385785fca9017c899; identity checks (patient, examination, eye, image, frame size): passed.
Measured run time 54.3 s, peak memory 2990 MiB.

5. Limitations

- Research prototype. No clinical validation; not a medical device.
- Stage III is the mean of the five official LG-DRG folds (research use, CC BY-NC 4.0 weights).
- No licensed or clinically validated image set has been used for acceptance.
- Only the CPU execution provider has been exercised; CUDA is not verified.
- Model weights are CC BY-NC 4.0 (non-commercial use only).
- Segmentation masks and the source image are not embedded; the only picture is the attribution overlay below.
- Attribution is influence evidence for the stated target only; it is not a lesion segmentation or a diagnosis.

6. AI Explainability (research only)

RESEARCH ONLY - NOT A DIAGNOSIS

Method: Occlusion sensitivity
Target: Grade 3 probability, mean of five folds
Model scope: Stage III five-fold ensemble (folds [0, 1, 2, 3, 4])
Baseline target score: 0.762148
Raw map range: -0.00738 to +0.08200 (positive support: true, negative support: true)
Forward calls: 325; wall time 322.6 s; provider Cpu
Source image SHA-256: 32d756fc923262daa5296c6852c1569e01394b37db57e72385785fca9017c899
Attribution map SHA-256: 229ece8be90c7326237a04614e8b2517f04d315ed1b9ce657709bca2dc787a63
Algorithm: retinacare-attribution-1.0; generated (unix) 1793203105
Model hierarchiretina-stage3-lgdrg-fold0: sha256 31f20d62f972a0a8f9119162c8a30b7e2a96509933f0a9bc9a3845b3a6129cb7
Model hierarchiretina-stage3-lgdrg-fold1: sha256 3f0deb84cb04a6c301cf73efdd154fe18297252430e7c3afa36d2ab087092620
Model hierarchiretina-stage3-lgdrg-fold2: sha256 a55301c7e27251047e10211dfb10ca7f73674c826d89c4e7a772f3476df0a3ab
Model hierarchiretina-stage3-lgdrg-fold3: sha256 ee7c16614859ef3c8e8d1c8fbcc5a07388e3bd02622f74bdedcd8c4772b6fcfe
Model hierarchiretina-stage3-lgdrg-fold4: sha256 659e65fb336342b2afc52385f4821a673e1b569b81454798820f4798fe876238
Parameter baseline_value: 0
Parameter grid: square_patches_constant_fill
Parameter masks: held fixed (RGB influence only)
Parameter patch: 64
Parameter perturbed_channels: rgb_only
Parameter perturbed_score_max: 0.7695231633260846
Parameter perturbed_score_min: 0.680150527949445
Parameter stride: 64
Parameter target_score: mean 5-way probability of the folds in scope
Validity checks passed: finite values, non-constant map, map hash, binding to patient, examination, run, eye and image hash.
Attribution shows which image regions influenced the selected model score. It is not a lesion segmentation, not a diagnosis and not proof of pathology. RESEARCH ONLY - NOT A DIAGNOSIS.

Attribution overlay: Occlusion sensitivity / Grade 3 probability, mean of five folds / Stage III five-fold ensemble (opacity 55%). RESEARCH ONLY - NOT A DIAGNOSIS.

